

Predicting F1 Race Winners through Data Curation: Metadata & Data Dictionary

Introduction

This data dictionary serves as a structured reference document of the data that we are curating. It ensures that every individual who interfaces with our results can understand the variables in a clear manner. In this project, the many tables correspond to different fields of information that can be used in the process to predict Formula 1 race winners.

Metadata

```
JSON
{
  "@context": "https://schema.org",
  "@type": "Dataset",
  "name": "Predicting F1 Race Winners through Data Curation",
  "description": "A curated dataset of Formula 1 sessions, drivers, circuits, weather conditions, tyre compounds, and race outcomes designed to support the prediction of Formula 1 race winners.",
  "creator": [
    {
      "@type": "Person",
      "name": "Fatih Atlamaz"
    },
    {
      "@type": "Person",
      "name": "Navtej Kathuria"
    },
    {
      "@type": "Person",
      "name": "Kunal Daftari"
    }
  ],
  "publisher": {
    "@type": "Organization",
    "name": "University of Illinois Urbana-Champaign"
  },
  "inLanguage": "en",
  "license": "https://creativecommons.org/licenses/by-nc/4.0/",
  "temporalCoverage": "2023/2025",
  "spatialCoverage": {
    "@type": "Place",
```

```

    "name": "Global Formula 1 Circuits"
  },
  "distribution": {
    "@type": "DataDownload",
    "encodingFormat": [
      "text/csv",
      "application/sql"
    ]
  },
  "version": "1.0",
  "dateCreated": "2025-11-10"
}

```

Sessions

```

SQL
create table sessions (
  session_key INTEGER PRIMARY KEY,
  session_type TEXT NOT NULL,
  session_name TEXT NOT NULL,
  circuit_key INTEGER NOT NULL REFERENCES circuits(circuit_key),
  date_start DATE NOT NULL,
  date_end DATE NOT NULL,
  year INTEGER NOT NULL
);

```

Sessions - represents the data needed to know what part of the 3-day race weekend the data is referring to (Free Practice 1, Free Practice 2, Qualifying, Race, or Sprint).

- Session Key - Unique ID to reference the entry
- Session Name and Session Type - Informal and Formal Name for Race type
- Date Start, Date End, Year - Identifies which year and when in the season a race event occurs
- Circuit Key - Identifies which of the F1 circuits the data references

Starting Grid

SQL

```
create table starting_grid (  
    session_key INTEGER NOT NULL REFERENCES sessions(session_key),  
    driver_number INTEGER NOT NULL REFERENCES drivers(driver_number),  
    position INTEGER NOT NULL,  
    lap_duration FLOAT NOT NULL,  
  
    PRIMARY KEY (session_key, driver_number)  
);
```

Starting Grid - Data to inform on the order in which the cars line up at the start of the race (based on data from the qualifying event the day prior)

- Session Key - Unique ID to reference the entry
- Driver Number - Driver selected (or 1, denoting previous WDC) number on the racecar (unique identifier)
- Position - Order on the grid (typically 1-20) sometimes can be from the pitlane (special situations denoted)
- Lap Duration - The qualifying lap time (duration) in which the driver achieved their position

Race Result

SQL

```
create table race_result (  
    session_key INTEGER NOT NULL REFERENCES sessions(session_key),  
    driver_number INTEGER NOT NULL REFERENCES drivers(driver_number),  
    position INTEGER NOT NULL,  
    driver_status TEXT NOT NULL CHECK (driver_status IN ("dnf", "dns", "dsq",  
"finished")),  
  
    PRIMARY KEY (session_key, driver_number)  
)
```

- Race Result - General data provided to identify the race result for a given session key; Tells find resulting positions per driver, as well as the potential flags attached with the race results
- Session Key - Unique ID to reference the entry
- Driver Number - Driver selected (or 1, denoting previous WDC) number on the racecar (unique identifier)
- Position - Where in the field the drivers ended up, depends on the classification (practice, quali, race, etc).

- Driver Status - the outcome of the race per driver (DNF, DNS, DSQ, Finished), tells the final resulting flags

Circuits

```
SQL
create table circuits (
  circuit_key INTEGER PRIMARY KEY,
  country_name TEXT NOT NULL,
  country_code TEXT NOT NULL
);
```

Circuits - Serves primarily as a reference table to provide basic information about the different F1 circuits and the basic geographic details about race.

- Circuit Key - Unique identifier to each circuit (integer only)
- Country Name- Full Country Name (Text)
- Country Code - International Standard 3-letter Country Identifiers

Drivers

```
SQL
create table drivers (
  session_key INTEGER NOT NULL REFERENCES sessions(session_key),
  driver_number INTEGER PRIMARY KEY,
  first_name TEXT NOT NULL,
  last_name TEXT NOT NULL,
  team_name TEXT NOT NULL REFERENCES teams(team_name),

  PRIMARY KEY (session_key, driver_number)
);
```

Drivers - All personally identifiable information related to the drivers on the grid in relation to F1

- Session Key - Unique identifier (same as ones referenced above)
- Driver Number - Driver selected (or 1, denoting previous WDC) number on the racecar (unique identifier)
- First Name and Last Name - Self explanatory, first and last name of the driver
- Team Name - One of the 10 teams that has cars on the grid in Formula 1

Teams

```
SQL
create table teams (
    team_name TEXT PRIMARY KEY
);
```

Teams - Very simple table simply used a reference table for other tables to search through teams on the grid by name

- Team Name - 10 teams that are on the Formula 1 Grid

Race Stints

```
SQL
create table race_stints (
    session_key INTEGER NOT NULL REFERENCES sessions(session_key),
    driver_number INTEGER NOT NULL REFERENCES drivers(driver_number),
    lap_start INTEGER NOT NULL,
    lap_end INTEGER NOT NULL,
    compound TEXT NOT NULL CHECK (compound IN ("SOFT", "MEDIUM", "HARD")),
    tyre_life INTEGER NOT NULL

    PRIMARY KEY (session_key, driver_number)
);
```

Race Stints -

- Session Key - Unique identifier (same as ones referenced above)
- Driver Number - Driver selected (or 1, denoting previous WDC) number on the racecar (unique identifier)
- Lap Start/End - When the specific lap in the stint began and ended, serves as a lap interval measure
- Compound - Which compound was the lap driven on (Soft, Medium, Hard, Wet, Inters)
- Tyre Life - How many laps old the tyre is already been used for in the current grand prix

Overtakes

```
SQL
create table overtakes (
```

```

circuit_key INTEGER NOT NULL REFERENCES circuits(circuit_key),
driver_number INTEGER NOT NULL REFERENCES drivers(driver_number),
overtake_avg FLOAT

PRIMARY KEY (driver_number, circuit_key)
);

```

Overtakes - Key statistics related to each driver and the number of overtakes and what the structure of the overtake was during the race (overtaking driver number, over-taken driver number, and position which the driver passing takes after the move)

- Circuit Key - Unique identifier to each circuit (integer only)
- Driver Number - Driver selected (or 1, denoting previous WDC) number on the racecar (unique identifier)
- Overtake Average - [Fatih Atlamaz](#) Idk where this came from lmfao?

Weather

```

SQL
create table weather (
  session_key INTEGER NOT NULL REFERENCES sessions(session_key),
  circuit_key INTEGER NOT NULL REFERENCES circuits(circuit_key),

  temperature FLOAT,
  humidity FLOAT,
  rainfall BOOLEAN,
  track_temperature FLOAT,
  wind_speed FLOAT,
  wind_direction FLOAT

  PRIMARY KEY (session_key, circuit_key)
);

```

Weather - Weather data related to each race, with multiple different metrics as well as track conditions which can (and typically do) greatly influence the outcome of a race.

- Session Key - Unique identifier (same as ones referenced above)
- Circuit Key - Unique identifier for the circuit where the data is from (same one as referenced above)
- Temperature - Outside temperature (provided in degrees Celsius)
- Humidity - Outside city humidity (relative humidity percentage)
- Rainfall - Outside city rainfall during the session (true/false)

- Track Temperature - Physical temperature of the track identified in the circuit key
- Wind Speed - Outside city wind speed (provided in meters per second unit)
- Wind Direction - Cardinal direction in which the wind is blowing at the circuit

Pirelli Tyres

SQL

```
create table pirelli_tyres (
  circuit_key INTEGER NOT NULL REFERENCES circuits(circuit_key),
  tyre_compound TEXT NOT NULL CHECK (tyre_compound IN ('C1', 'C2', 'C3',
'C4', 'C5', 'C6')),
  year INTEGER NOT NULL,

  PRIMARY KEY (circuit_key, year)
);
```

Pirelli Tyres - Provides the data on the tires available and/or used during a specific stint on the race

- Circuits Key - Identifies which of the F1 circuits the data references
- Tyre Compound - From C1 through C6, which represent the different levels of tyre thickness and tread (from the following: Soft, Medium, Hard, Inters, Wets)
- Year - Year in which the specific data is related and relevant to